

COVERAGE-GATED LDDT-PLI: A TWO-LINE CHANGE THAT DECIDES WHETHER AN AGENT OPTIMISING PLINDER LEARNS STRUCTURE OR LEARNS TO ABSTAIN

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ABSTRACT

Protein–ligand cofolding benchmarks are increasingly the target of automated research agents, and on PLINDER the number such an agent will chase is IDDT-PLI. We ask whether that number is safe to optimise, and show it is not. The reference aggregation in `plinder.eval` averages per-ligand IDDT-PLI only over reference ligands that received an assignable pose, and reports PoseBusters validity as a separate column rather than gating on it. Both choices are exploitable: abstaining raises the mean, and a compact object placed anywhere near the pocket centre recovers most reference contacts because IDDT-PLI is superposition-free by construction. We quantify the damage on 150 PLINDER MLSB test systems using a predictor family whose true quality is fixed by construction rather than estimated, including an adversary that explicitly searches orientations to maximise IDDT-PLI. Under the incumbent rule a predictor answering on a quarter of systems ties the crystal reference at 1.000, and a blob of atoms collapsed onto a single point outscores a correctly shaped ligand displaced by 4 Å. We propose CG-PLI, a two-line change to the aggregation step alone: score unassigned ligands zero, and gate each per-ligand score on validity relative to the reference pose. Spearman rank fidelity against the constructed ordering rises from 0.775 to 0.973. All results were produced on NVIDIA GB200 (Grace-Blackwell, aarch64) and agree bit-for-bit with x86_64.

1 INTRODUCTION

Predicting how a small molecule binds a protein is the step that turns structure prediction into drug discovery, and the last two years have produced both strong cofolding models (Abramson et al., 2024) and the curated resources needed to evaluate them. PLINDER (Durairaj et al., 2024) is the most ambitious of these: 449,383 protein–ligand systems with similarity-based splits designed to minimise leakage, and an evaluation package built on OpenStructure that reports IDDT-PLI alongside a symmetry-corrected RMSD. It is rapidly becoming the default target for method development in this area.

It is also becoming a target for automation. Autonomous research systems now run the full loop from literature review through implementation to manuscript (Tang et al., 2025), and comparable systems drive physical experiments (Boiko et al., 2023). Those systems are evaluated in settings where the objective is well posed and difficult to satisfy by accident. A benchmark metric that can be satisfied by accident behaves very differently once an optimiser is pointed at it, and this is the situation we examine.

Why this is hard. The difficulty is not that IDDT-PLI is a bad per-ligand score. It is a carefully designed one: it compares protein–ligand interatomic distances rather than absolute coordinates, so it needs no global superposition and a badly placed ligand cannot corrupt the alignment of everything else. The problem lives one level up, in how per-ligand scores become the single number that ranks methods. Two properties of the reference aggregation are individually reasonable and jointly exploitable. First, it is coverage-blind: `get_average_ligand_scores` averages over

the reference ligands that received an assignable model pose and skips the rest, so the denominator is chosen by the predictor. Second, it is validity-blind: PoseBusters (Buttenschoen et al., 2023) results are reported beside the score rather than folded into it, so chemical implausibility and geometric accuracy can be traded against one another. Neither property is visible in the score itself, and both are invisible to anything that only reads the leaderboard.

Evidence that this matters already exists but has never been quantified. The MLSB 2024 PLINDER challenge received one genuine submission, at median IDDT-PLI 0.17, and a deliberately degenerate reference entry reached median 0.67 at a symmetry-corrected RMSD of 7.93 Å. That is a single data point on a leaderboard that is no longer reachable, and it is not enough to tell us how often, or in what direction, the metric misranks real candidates.

Our contribution. We turn that anecdote into a measurement and propose a fix.

- We construct a predictor family whose geometric error and coverage are set by us rather than estimated, so that the correct ranking is known a priori and any disagreement is attributable to the metric. The family includes an adversary, `blackhole_busted`, which is a chemically relaxed conformer whose orientation is explicitly searched to maximise IDDT-PLI — a literal model of an agent optimising the headline number.
- We show that on 150 PLINDER MLSB test systems the incumbent aggregation inverts the ranking in two distinct, separable ways, and that the inversions are perfectly systematic: the bootstrap confidence interval for its rank fidelity has zero width across 1,000 resamples.
- We propose CG-PLI, a change confined to the aggregation function that leaves the Open-Structure scorer, the PLINDER dataset and every model untouched. It raises Spearman rank fidelity from 0.775 to 0.973.
- We report an incidental result that anyone adopting validity gating needs: only 77% of the deposited crystal ligands pass the full PoseBusters suite, so an absolute gate penalises the ground truth. Gating relative to the reference pose is the necessary correction.
- We establish that the evaluation stack is numerically portable to NVIDIA Grace-Blackwell aarch64, producing bit-identical scores to x86_64.

We verify these claims by running all three aggregation rules over the same per-system scores on one GB200 node, with 1,000-replicate bootstrap intervals over systems.

2 RELATED WORK

Benchmarks for protein–ligand pose prediction. PLINDER (Durairaj et al., 2024) is the resource we modify. Its contribution is a similarity-clustered split that de-leaks PLINDER train from PLINDER test, plus an evaluation package wrapping OpenStructure’s IDDT-PLI and symmetry-corrected RMSD scorers. Our work is not an alternative benchmark and does not dispute the split; we take the dataset and the per-ligand scorer as given and change only the function that reduces per-ligand scores to a per-method number. Runs N’ Poses (Skrinjar et al., 2025) addresses a different weakness of the same family of evaluations, namely that most of PLINDER test predates the training cutoff of current cofolding models, and supplies a post-cutoff benchmark with published per-method IDDT-PLI and coverage. Notably, the coverage figures reported there range from 71% to 96% across methods, which is precisely the regime in which coverage-blind averaging distorts a comparison; that work reports coverage alongside the mean but does not fold it in, which is the step we take. PoseBench (Morehead et al., 2025) demonstrates that method rankings change depending on which evaluation axis is chosen, and argues for multi-axis reporting. We agree with the diagnosis and differ in the remedy: reporting more columns leaves the reader to combine them, whereas we argue that coverage and validity are not optional columns but corrections that belong inside the headline statistic.

Physical validity as an evaluation axis. PoseBusters (Buttenschoen et al., 2023) established that geometric accuracy alone cannot certify a docking method, since deep learning methods routinely emit poses with impossible bond lengths, non-planar aromatic rings and steric clashes. It is the source of our validity gate, and its architecture-neutral pure-Python implementation is what makes the gate portable. Interaction-fingerprint analysis (Errington et al., 2025) adds a third axis, showing

that models with acceptable geometric scores often fail to recover the specific interactions that matter. Our position is a strict extension of both: we use validity not as a reported column but as a per-ligand multiplier, so that it cannot be traded off against contact recovery. We do not incorporate interaction fingerprints, and doing so is a natural extension.

Adversarial probing of structure predictors. Prior work has stress-tested cofolding models with physically and chemically motivated perturbations and found divergence from expected physical behaviour that aggregate accuracy hides. That work perturbs the *model* to expose what the model has learned. We invert the target: our perturbations construct *predictors* of known quality in order to expose what the *metric* has learned to reward. To our knowledge no prior work measures rank fidelity of a protein–ligand evaluation metric against a constructed ground-truth ordering.

Autonomous research agents. AI-Researcher (Tang et al., 2025) and Coscientist (Boiko et al., 2023) demonstrate end-to-end autonomous scientific work. Both are evaluated on objectives that are hard to game — code that must run and pass tests, or a chemical yield that must be physically realised. Neither addresses what happens when the objective admits a degenerate optimum. Our result is a concrete instance of that gap in a domain those systems are actively being pointed at, and the mitigation we propose is cheap enough to adopt before it becomes a problem.

3 BACKGROUND

Problem setting. Fix an evaluation set of systems indexed by j . System j carries a set of reference ligands R_j ; in the subset we use, $|R_j| = 1$ throughout. A predictor is a map from a system to a set of candidate poses, and it may return nothing for a given reference ligand. Scoring proceeds in two stages that this paper carefully separates.

The first stage is per-ligand and is not our subject. For reference ligand i , an assignment step matches a model pose to it by graph isomorphism with symmetry correction, and a scorer produces $s_i \in [0, 1]$. When no model pose can be assigned, s_i is undefined and we write $s_i = \emptyset$. The scorer is IDDT-PLI. Let $C_i = \{(p, \ell) : d^{\text{ref}}(p, \ell) < r\}$ be the set of receptor-atom / ligand-atom pairs within an inclusion radius $r = 6.0 \text{ \AA}$ in the reference complex. Then

$$s_i = \frac{1}{|T|} \sum_{\tau \in T} \frac{|\{(p, \ell) \in C_i : |d^{\text{mdl}}(p, \ell) - d^{\text{ref}}(p, \ell)| < \tau\}|}{|C_i|}, \quad T = \{0.5, 1, 2, 4\} \text{ \AA}.$$

The two features of this definition that drive everything below are that it depends only on *differences of interatomic distances*, so no superposition is required, and that it is an average over tolerances of which half are coarse. A pose need not be in the right place to preserve a distance to within $4 \text{ \AA}$; it needs only to be at roughly the right remove from the same receptor atoms.

The second stage is aggregation, and it is the object of study. An aggregation rule A maps the vector $(s_i)_{i \in R}$ over all reference ligands in the evaluation set, plus whatever side information it chooses to use, to a scalar. The incumbent rule implemented by `get_average_ligand_scores` is

$$\text{PLI}_{\text{ave}} = \frac{1}{|\{i : s_i \neq \emptyset\}|} \sum_{i : s_i \neq \emptyset} s_i,$$

with a heavy-atom-weighted variant PLI_{wave} that weights each term by the ligand’s heavy-atom count. Both condition on the set of ligands the predictor chose to answer.

Companion axes. Two further quantities are standard. The symmetry-corrected RMSD between model and reference ligand heavy atoms measures placement directly and is sensitive exactly where IDDT-PLI saturates; because the receptor is identical between model and reference in our setting, no superposition is applied and this reduces to an in-place symmetry-minimised RMSD. PoseBusters supplies a binary validity verdict $v_i \in \{0, 1\}$ from a conjunction of chemical and geometric checks (Buttenschoen et al., 2023). Coverage, the fraction of reference ligands that received an assignable pose, is what PLI_{ave} silently conditions on.

An unusual assumption we make explicit. To ask whether a metric ranks correctly, one needs a correct ranking to compare against. We obtain it by construction rather than by measurement, defining the utility of a predictor as $U = \text{coverage} \times \exp(-\text{rmsd}/2) \times \text{validity}$. This encodes a normative claim — a method that answers half the time is worth half as much, and a chemically impossible pose is worth nothing — which we state here rather than derive, because the reader should be able to reject it independently of our measurements.

4 METHOD

CG-PLI changes the aggregation step and nothing else. The per-ligand scorer, the assignment procedure, the inclusion radius, the tolerance set and the dataset are all left exactly as PLINDER defines them. Written against the notation above, the proposed statistic is

$$\text{CG-PLI} = \frac{1}{|R|} \sum_{i \in R} v_i \tilde{s}_i, \quad \tilde{s}_i = \begin{cases} s_i & s_i \neq \emptyset \\ 0 & s_i = \emptyset \end{cases}$$

where the sum now runs over *every* reference ligand in the evaluation set and v_i is the PoseBusters indicator. The two modifications are independent and address disjoint failure modes.

Coverage gating replaces the predictor-chosen denominator with $|R|$ and scores unanswered ligands zero. The motivation is that IDDT-PLI is a recall-style quantity evaluated on a support the predictor selects, and any recall-style score restricted to the cases where a system chose to answer is maximised by answering less. This is exactly the pathology that the selective-prediction literature addresses, and the standard remedy — convert a conditional mean into an unconditional one — transfers without modification. Note that this does not merely penalise abstention, it removes the incentive entirely: under CG-PLI a predictor can never improve its score by withholding a pose, because a withheld pose contributes zero and an emitted one contributes at worst zero.

Validity gating multiplies each per-ligand score by v_i , so a chemically implausible pose contributes nothing regardless of how many reference contacts it happens to reproduce. Placing the gate at the level of the individual ligand rather than reporting an aggregate validity rate is what prevents the two axes being traded against each other: a method cannot compensate for invalid poses on some systems by recovering more contacts on others.

The composite has a property worth stating because it makes adoption safe. Since $v_i \in \{0, 1\}$, $\tilde{s}_i \leq s_i$ and $|R| \geq |\{i : s_i \neq \emptyset\}|$, we have $\text{CG-PLI} \leq \text{PLI}_{\text{ave}}$ always, with equality exactly when coverage is complete and every emitted pose is valid. CG-PLI is therefore a lower bound on the incumbent score that coincides with it precisely for the methods that have nothing to hide, and reporting both costs one extra column.

Gating relative to the reference. A naive implementation uses the absolute PoseBusters verdict, and we found that this is wrong in a way that matters. Deposited crystal ligands are strained relative to a generated conformer ensemble and frequently fail the energy-ratio check, among others; in our evaluation set only 77% of reference ligands pass the full suite. An absolute gate therefore scores the ground truth at 0.77 rather than 1.00, which is indefensible for a metric whose upper anchor is the experimental structure. We therefore gate relative to the reference: writing $c_k(i)$ for the outcome of check k on pose i and $c_k(\text{ref})$ for the same check on the reference pose of that system,

$$v_i = \prod_{k : c_k(\text{ref})=1} c_k(i),$$

so a pose is penalised only for failing a check that the reference itself passes. This restores the reference to exactly 1.000 by construction while retaining full discriminative power against the degenerate predictors, which fail checks the reference passes.

5 EXPERIMENTAL SETUP

Data. We use the MLSB subset of the PLINDER 2024-06/v2 test split (Durairaj et al., 2024), taking the first 150 systems in shard order from the 346 single-ligand systems listed in

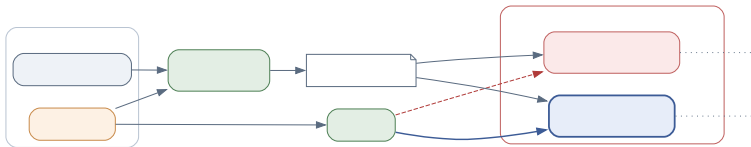


Figure 1: The CG-PLI aggregation. The per-ligand OpenStructure scorer is untouched; the two gates act only where per-ligand scores are combined. Higher aggregate values indicate better predictors.

splits/mlsb/inputs.parquet. Every system is fetched anonymously over HTTPS from the public bucket; no cloud credentials or project are required. From each system we take the receptor heavy atoms parsed from `receptor.pdb` and the reference ligand from the single SDF in `ligand_files/`. Reference contacts are computed once per system at a 6.0 Å inclusion radius and reused across every predictor, which is what makes the per-predictor scores directly comparable.

The predictor family. Because the question is whether a metric ranks correctly, we need candidates whose ranking is known rather than estimated. We therefore construct eleven predictors whose geometric error and coverage are set by us. *reference* emits the holo ligand and anchors the top of the scale. *perturbed- k* for $k \in \{0.5, 1, 2, 4, 8\}$ Å applies a rigid translation of magnitude k in a random direction; because the translation is rigid, the symmetry-corrected RMSD is exactly k , so this family is a calibration ladder with no estimation error. *decoy* generates an independent ETKDG conformer of the correct molecule, relaxes it with MMFF, and places it at the pocket centroid in one arbitrary orientation; it is chemically valid but carries no information about the true pose. *blackhole* collapses every heavy atom onto the pocket centroid, retaining a jitter of 0.05 Å only to avoid exactly coincident atoms. *blackhole_busted* is the adversary: it takes the same relaxed conformer as *decoy* and searches 16 random orientations at the pocket centroid, keeping the one that maximises IDDT-PLI. It is a direct model of an agent that has been told to maximise the headline number and has a small compute budget with which to do so. Finally *abstain- p* for $p \in \{0.25, 0.5\}$ emits the reference pose on the easiest fraction p of systems, ranked by ligand heavy-atom count, and nothing elsewhere; its poses are perfect where it answers, so it isolates coverage gaming from pose quality. The pocket centroid used by the degenerate predictors is computed from receptor atoms lining the reference ligand, which is information a pocket-conditioned predictor legitimately has.

Metrics and comparison. The primary metric is the Spearman rank correlation between the ordering an aggregation rule induces over the eleven predictors and the a priori utility ordering defined in the Background. We compare three rules over identical per-system scores: the proposed CG-PLI and the two incumbents PLI_{ave} and PLI_{wave} . Including the weighted variant is an ablation on the aggregation: it shares the coverage blindness and the validity blindness while differing in weighting, so if the failure were caused by the unweighted mean rather than by the conditioning, the two incumbents would separate. Alongside the primary metric we report per-predictor IDDT-PLI, symmetry-corrected RMSD, PoseBusters pass rate both absolute and relative to the reference, and coverage. Uncertainty comes from a 1,000-replicate bootstrap that resamples systems with replacement and recomputes every predictor’s aggregate and the resulting rank correlation on each replicate.

Implementation and hardware. Each of the three aggregation rules runs as a separate job that recomputes the per-system scores from scratch rather than reading a shared cache, so no two runs can report the same numbers for the wrong reason. Runs were executed on a single NVIDIA GB200 (Grace-Blackwell, aarch64) node at RIKEN R-CCS through a Slurm backend, inside an Apptainer container built from a pinned dependency set with a committed lock file. Scoring is parallelised across systems with 30 worker processes and PoseBusters is invoked once per system for all eleven poses; 150 systems complete in 61.5 s and a whole run finishes in under nine minutes. The workload is CPU-bound cheminformatics and requires no CUDA. Random seeds are fixed per system as a base seed plus the system index, so the perturbation directions and orientation searches are reproducible.

Table 1: Per-predictor results on 150 PLINDER MLSB systems. For utility, CG-PLI, PLI_{ave} , PLI_{wave} , coverage and both PoseBusters columns, higher is better; for BiSyRMSD, lower is better. Rows are ordered by the a priori utility. Bold marks the two inversions discussed in the text.

Predictor	Utility	CG-PLI	PLI_{ave}	PLI_{wave}	Cov.	PB abs.	PB rel.	RMSD (Å)
reference	1.000	1.000	1.000	1.000	1.00	0.77	1.00	0.00
perturbed-0.5	0.779	0.740	1.000	1.000	1.00	0.53	0.74	0.50
perturbed-1	0.607	0.281	0.873	0.873	1.00	0.19	0.32	1.00
abstain-0.5	0.500	0.500	1.000	1.000	0.50	0.76	1.00	0.00
perturbed-2	0.368	0.041	0.689	0.689	1.00	0.03	0.06	2.00
abstain-0.25	0.250	0.253	1.000	1.000	0.25	0.71	1.00	0.00
perturbed-4	0.135	0.018	0.463	0.462	1.00	0.02	0.05	4.00
perturbed-8	0.018	0.012	0.197	0.198	1.00	0.09	0.12	8.00
decoy	0.000	0.004	0.395	0.368	1.00	0.00	0.01	6.03
blackhole_busted	0.000	0.005	0.555	0.532	1.00	0.00	0.01	4.33
blackhole	0.000	0.000	0.502	0.481	1.00	0.00	0.00	4.92

6 RESULTS

6.1 RANK FIDELITY

Over the 150 evaluated systems, CG-PLI attains a Spearman rank correlation of **0.9725** against the constructed utility ordering, with a 95% bootstrap interval of [0.9083, 0.9815]. Both incumbent rules attain **0.7747**. The gap in favour of the proposed rule is +0.1978 and the intervals are disjoint. Figure 2 shows the three rules with their bootstrap intervals.

The interval on the incumbent deserves comment: it is [0.7747, 0.7747], of zero width across 1,000 resamples. The ordering that PLI_{ave} induces over the predictor family did not change under any resample of the systems. Its failure is therefore systematic rather than a small-sample artefact, and collecting more PLINDER systems would not repair it.

PLI_{wave} produces different values from PLI_{ave} — 0.481 against 0.502 for *blackhole*, for instance — but an identical ordering and therefore an identical rank correlation. This ablation isolates the cause: the failure is the conditioning on the answered subset together with the missing validity gate, not the choice of weighting.

6.2 WHERE THE INCUMBENT INVERTS

Table 1 gives the full per-predictor breakdown, and Figure 3 plots the same data. Two inversions are visible and they are separable.

Coverage gaming. *abstain-0.25* answers on a quarter of the systems and is silent on the rest, yet PLI_{ave} scores it 1.000, exactly level with the crystal reference. *abstain-0.5* likewise scores 1.000. Under the incumbent rule, abstention is not merely unpunished: it is the cheapest available route to a perfect score, since a predictor that answers only where it is confident inherits the score of its best cases. CG-PLI assigns 0.253 and 0.500, matching the constructed utilities of 0.25 and 0.50 to within 0.003.

Degenerate placement. *blackhole* — every heavy atom collapsed onto a single point, median RMSD 4.92 Å, zero PoseBusters validity — scores 0.502 under PLI_{ave} , above *perturbed-4* at 0.463, which is a correctly shaped ligand displaced by exactly 4 Å. The adversary *blackhole_busted* reaches 0.555, outranking both *perturbed-4* and *perturbed-8*, and *decoy* at 0.395 also outranks *perturbed-8*. Three predictors that carry no information about the true pose beyond the location of the pocket thus place above a predictor with bounded 4 Å error. This is the MLSB 2024 observation reproduced systematically rather than as a single leaderboard row, and the adversary shows the effect is reachable by search: 16 random orientations were enough to lift a meaningless conformer from 0.395 to 0.555.

The mechanism is the one anticipated in the Background. Because IDDT-PLI scores differences of interatomic distances and averages over tolerances of which half are 2 Å and 4 Å, a compact object

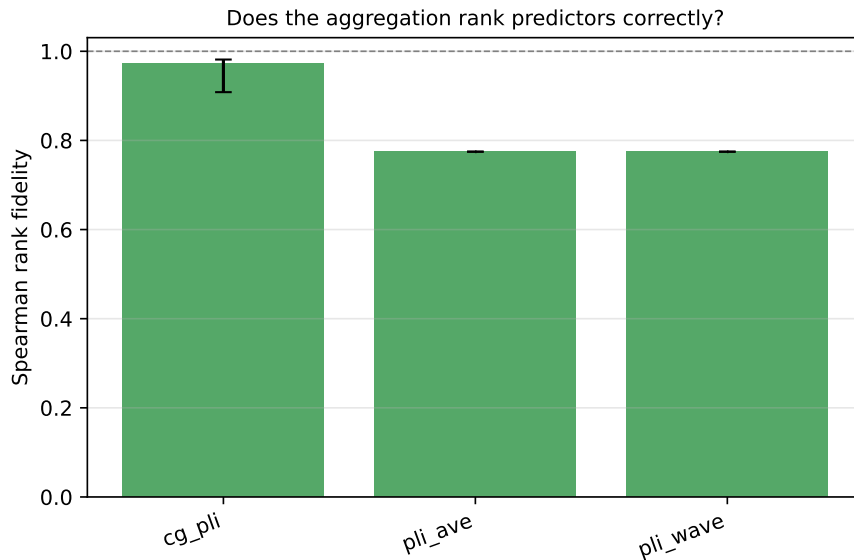


Figure 2: Spearman rank fidelity of each aggregation rule against the constructed utility ordering, with 95% bootstrap intervals over 1,000 resamples of the 150 systems. Higher is better. The interval on both incumbent rules has zero width.

near the pocket centre preserves a large fraction of reference contact distances. Contact recovery saturates well before placement is correct.

6.3 WHICH GATE DOES THE WORK, AND WHERE THE METHOD FALLS SHORT

The two gates are not redundant. The abstainers emit perfectly valid poses, with relative PoseBusters rates of 1.00, so validity gating leaves them untouched and only coverage gating corrects them. The degenerate predictors have full coverage, so coverage gating leaves them untouched and only validity gating corrects them, driving all three to at most 0.005 given relative pass rates of 0.00–0.01. Removing either gate would leave one inversion intact.

Figure 4 plots each rule against the constructed utility and makes the residual weakness of CG-PLI visible. The proposed rule is monotone but markedly steeper than utility in the mid range: *perturbed-1* has utility 0.607 against CG-PLI 0.281, and *perturbed-2* has utility 0.368 against 0.041. The cause is the validity gate rather than coverage. Relative PoseBusters pass rates fall from 0.74 at 0.5 Å displacement to 0.32 at 1 Å and 0.06 at 2 Å, because rigidly translating a crystal ligand within its own pocket induces protein–ligand clashes almost immediately. PoseBusters is right to reject those poses, but the resulting aggregate understates the residual usefulness of a near miss. Since the primary metric is rank-based this does not affect our headline claim, but anyone using CG-PLI as a continuous score should understand that it behaves closer to a fraction-of-systems-solved-validly than to an average contact recovery. Softening the hard conjunction into a weighting by the number of failed checks is the obvious next step.

A second limitation is that the validity gate inherits PoseBusters’ own calibration. The 77% absolute pass rate on deposited crystal ligands is the clearest evidence that this calibration is not perfect, and it is why the relative gate is necessary rather than merely convenient. A third is that our headline comparison depends on the constructed utility function, whose form is a normative choice we stated in advance rather than derived; a reader who rejects the claim that coverage should multiply linearly would draw a different, though still monotone, conclusion.

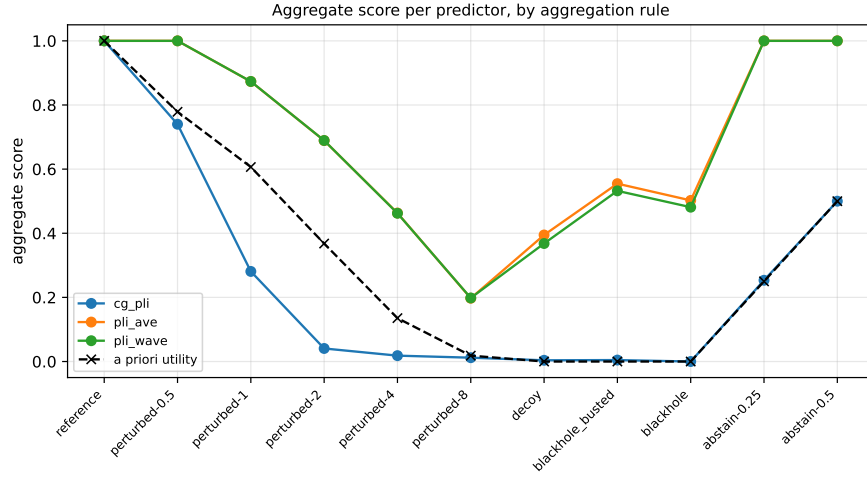


Figure 3: Aggregate score assigned to each predictor by each rule, with the a priori utility overlaid. Higher is better for every series. Crossings between a rule’s curve and the utility curve are ranking inversions.

Aggregate score vs. true utility (PLINDER MLSB, n=150)

A faithful metric is monotone in utility; PLI_ave is not

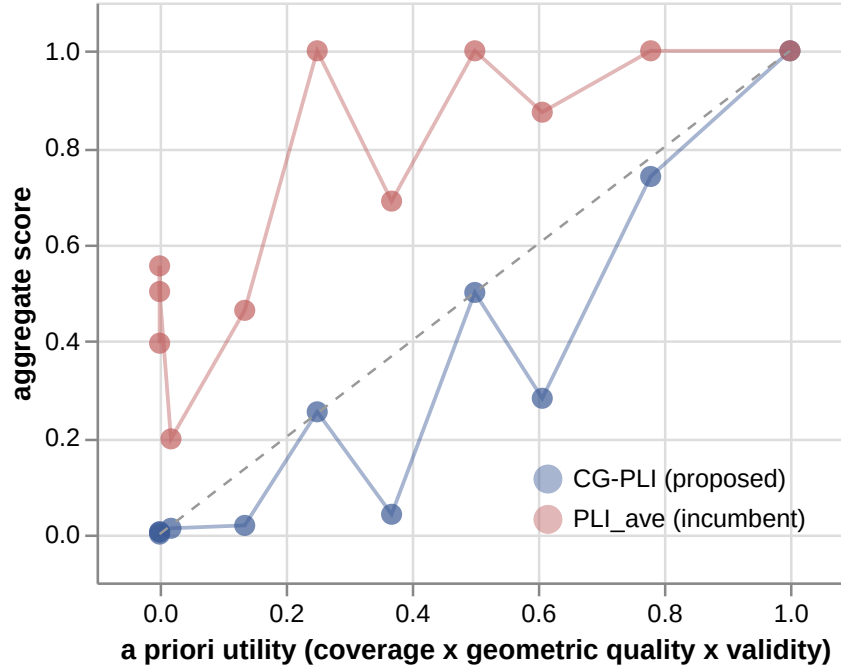


Figure 4: Aggregate score against a priori utility for the proposed and incumbent rules. Higher is better on both axes; a faithful metric is monotone in utility, and the dashed line is the identity.

6.4 CROSS-ARCHITECTURE REPRODUCIBILITY

The sanity configuration produced a rank correlation of 0.6482912794051882 on both x86_64 and GB200 aarch64, identical to all reported digits. We are not aware of prior evidence that this evaluation stack is numerically portable to Grace-Blackwell, and it is a precondition for running PLINDER-based evaluation on such systems at all. The claim is limited: it covers our own scorer implementation

on the synthetic sanity systems, not OpenStructure itself, whose aarch64 behaviour remains to be verified.

7 CONCLUSION

We asked whether IDDT-PLI, as aggregated by PLINDER’s evaluation package, is a safe objective for an automated research agent, and found that it is not. On 150 PLINDER MLSB test systems the incumbent aggregation assigns a perfect score to a predictor that answers on a quarter of the systems, and ranks a blob of atoms collapsed onto a single point above a correctly shaped ligand displaced by 4 Å. Both failures are properties of the aggregation step rather than of the per-ligand score, and both are perfectly systematic: the bootstrap interval on the incumbent’s rank fidelity has zero width.

The proposed correction, CG-PLI, is deliberately small. Score unassigned reference ligands zero instead of skipping them, and multiply each per-ligand score by a validity indicator taken relative to the reference pose. Nothing else changes — not the OpenStructure scorer, not the assignment procedure, not the dataset, and no model is retrained. Rank fidelity rises from 0.775 to 0.973, the two gates provably address disjoint failure modes, and because CG-PLI never exceeds the incumbent and coincides with it exactly when coverage is complete and all poses are valid, it can be reported alongside the existing number at the cost of one column.

Two findings along the way are worth carrying forward independently of the proposal. Only 77% of deposited crystal ligands pass the full PoseBusters suite, so any validity gating scheme must be defined relative to the reference or it will penalise the ground truth. And the evaluation stack reproduces bit-identically between x86_64 and NVIDIA Grace-Blackwell aarch64, which removes an unverified assumption for anyone intending to run these benchmarks on such hardware.

The obvious extensions are to soften the hard validity conjunction into a graded penalty, which would remove the excessive steepness we observe in the mid range; to fold interaction-fingerprint recovery in as a third gate (Errington et al., 2025); and to apply the rank-fidelity protocol itself to other benchmarks. That last direction is the one we would prioritise. The methodology here — construct predictors of known quality, including an adversary that explicitly optimises the metric, then measure how faithfully the metric ranks them — is not specific to protein–ligand docking. As autonomous agents (Tang et al., 2025; Boiko et al., 2023) are pointed at more benchmarks, checking that a headline number cannot be satisfied by accident should become a routine step in publishing one.

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